# One-Sample Proportion Inference

## Modeling a count of successes

Download the section 11.Rmd handout to `STAT240/lecture/sect11-single-proportion`.

Download the file `chimpanzee.csv` to `STAT240/data`.

Material in this section is covered by Chapter 12 on the notes website.

The chimpanzee data is based on an Emory University experiment.

- Chimpanzees choose from colored tokens
- One color is **selfish**, the other is **prosocial**
- Different combinations were studied

A chimpanzee can choose to act selfishly, or socially.

Modeling a binary population:

- Based on the binomial + normal
- $p$ is the unknown parameter of interest

(we saw this in chapter 10)

$p$ is the probability of a prosocial choice.

- Is $p_{partner}$ greater than 0.5?
- Is $p_{partner}$ the same as $p_{nopartner}$?

Let $X$ be the count of prosocial choices and $\hat{p}$ be the proportion of prosocial choices out of $n$.

$$X \sim Binom(n, p), \qquad \hat{p} = \frac{X}{n}$$

which relies on the BINS assumptions. $n$ is known and $p$ is the parameter of interest.

Chimpanzee A had 90 trials with a partner. Let's suppose $p = 0.6$.

Let's try a simulation of possible values of $\hat{p}$.

Recall that the normal is a good approximation to the binomial when $np(1-p)$ is large.

$$X \dot{\sim} N(np, \sqrt{np(1-p)})$$

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

- $E(\hat{p}) = p$
- Error decreases with $n$
- Error is a function of $p$

Inference for $p$ is based on the binomial and normal.

- CI critical values
- Null distribution for a hypothesis test

To use the normal, we like to have $np(1-p) \geq 10$.

Back to chimpanzee $A$. We want to study $p_{A,partner}$.

We pretended $p_{A,partner} = 0.6$ and simulated different values of $\hat{p}$.

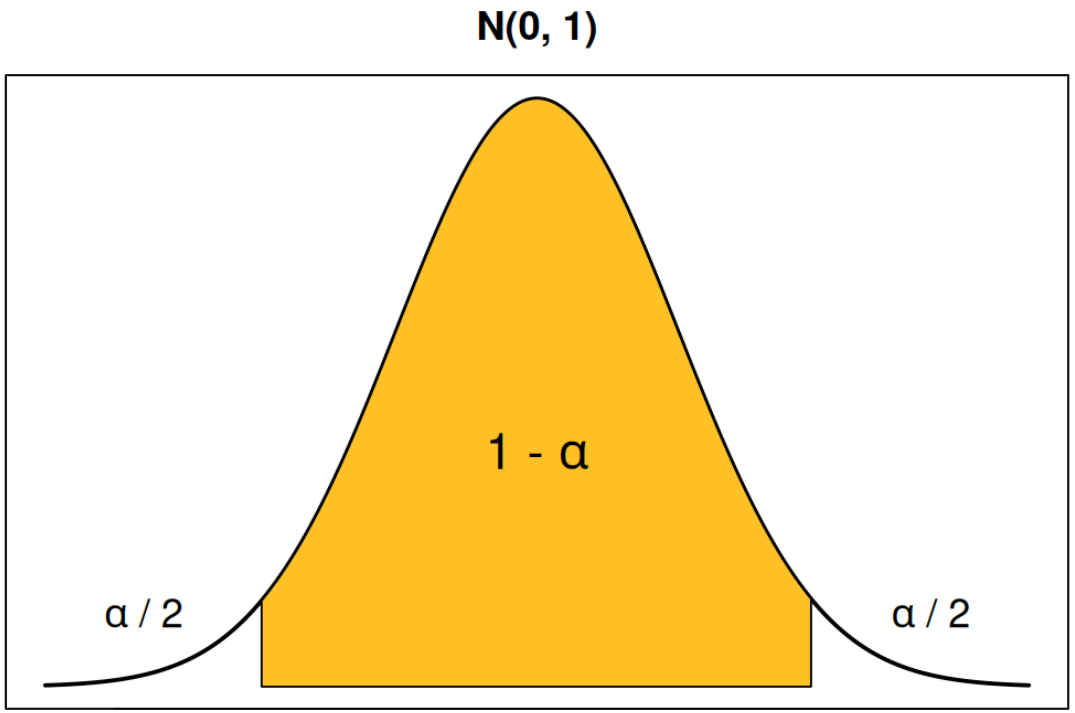
In practice, we don't have $p_{A,partner}$, just a single $\hat{p} = 60/90$.

CI formula:

$$\text{point estimate} \ \pm \ \text{critical value} \ \times \ \text{standard error}$$

- For $p$, the point estimate is $\hat{p}$
- The standard error is $\sqrt{\frac{p(1-p)}{n}}$
- The critical value is from $N(0, 1)$

Find a critical value with `qnorm()`.




For a 90% CI, use 1.645.

Problem: don't have $\sqrt{\frac{p(1-p)}{n}}$. Use $\hat{p}$ instead:

$$\hat{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

We are 90% confident that $p_{A,partner}$ is within (0.585, 0.748).

This interval is approximate.

Now, let's look at a hypothesis test approach for $p$. Two ways:

- Binomial null
- Normal null (Z test)

Both use BINS, and Z test also depends on $n$ and $p$.

Model the count of prosocial choices made by chimpanzee A (with a partner) as

$$Binom(90, p)$$

Is chimpanzee A more prosocial than selfish?

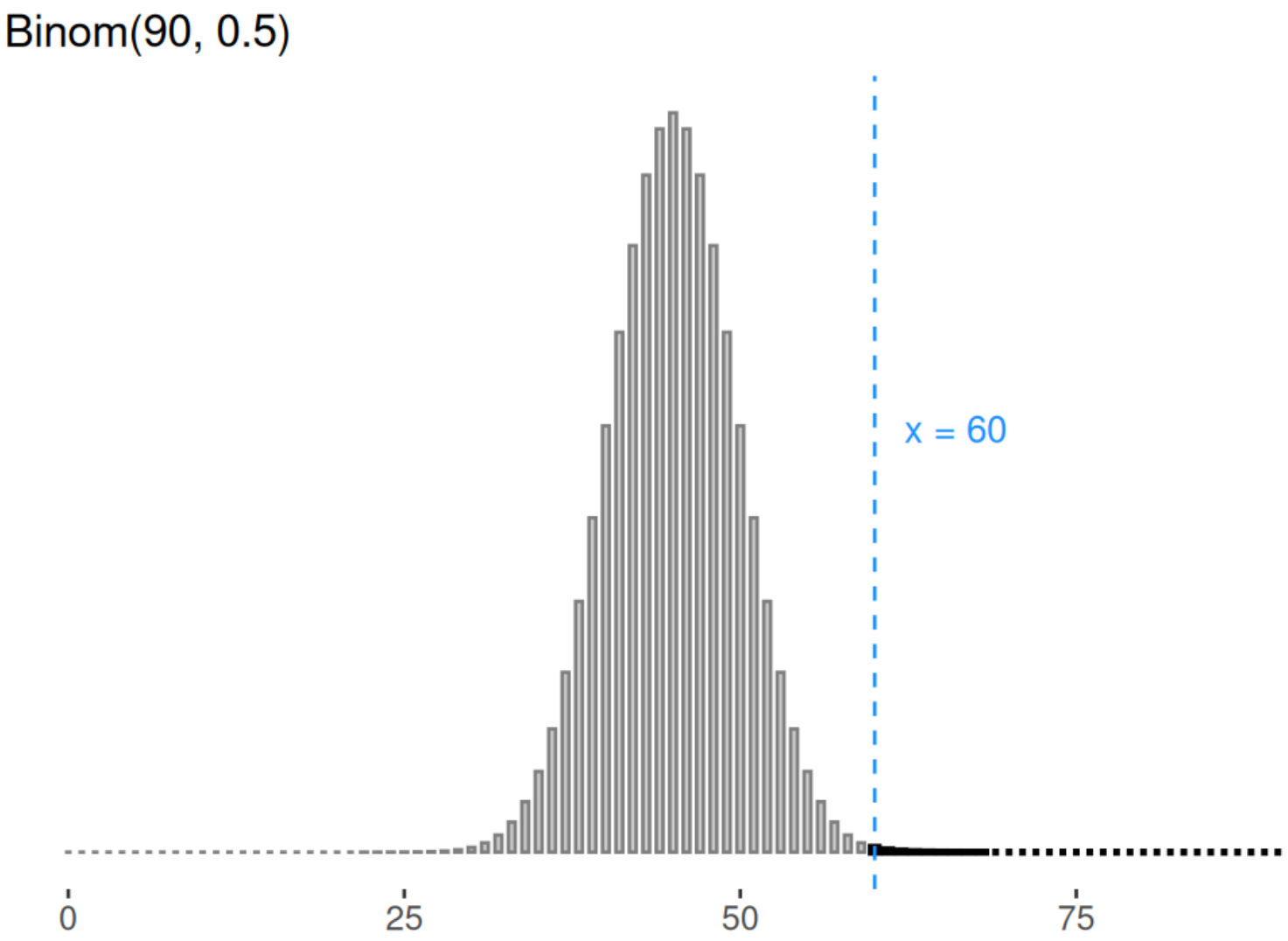
$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p > 0.5$$

Let's use $\alpha = 0.1$.

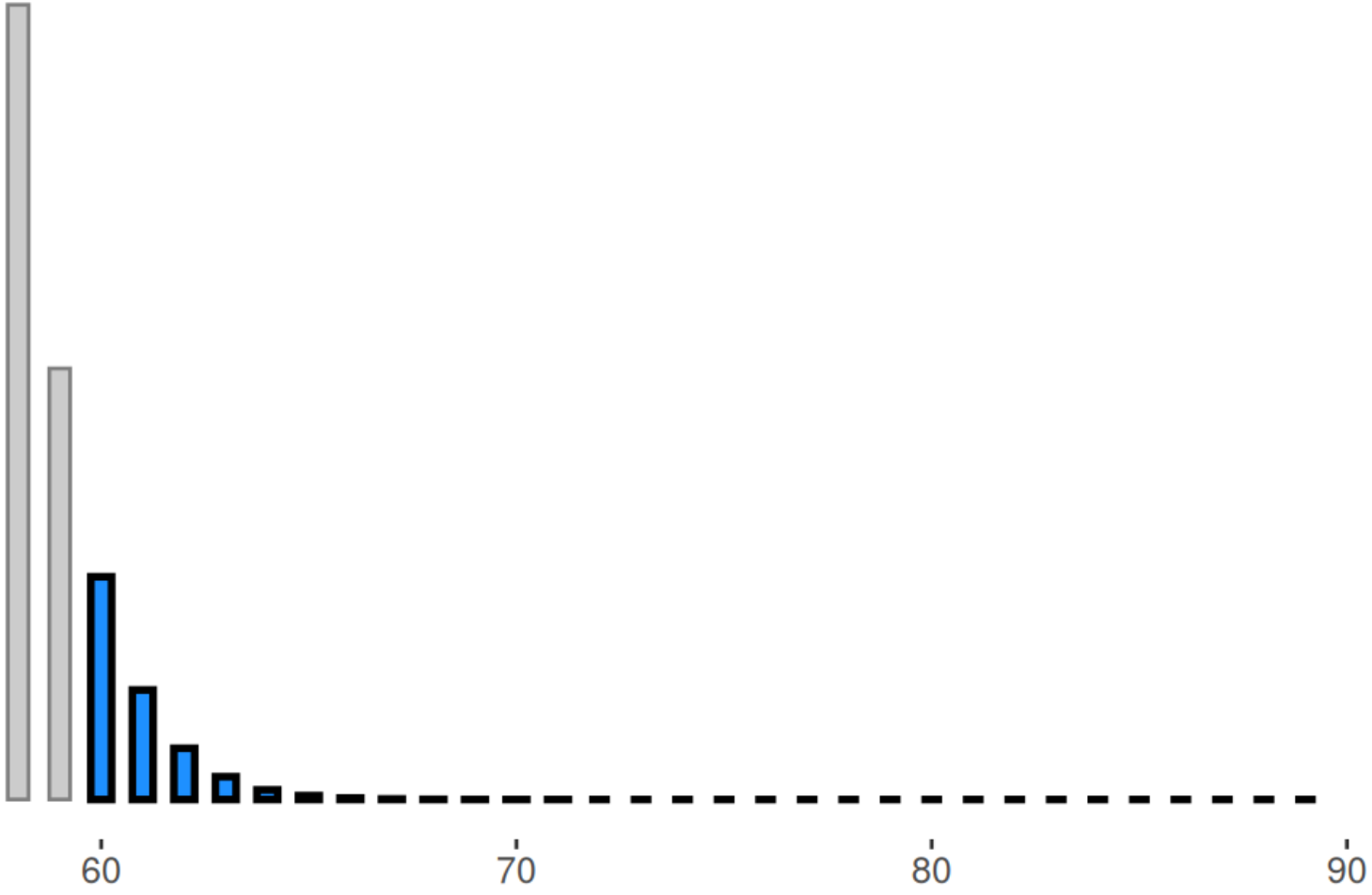
Our test statistic is $X$ has null distribution

$$Binom(90, 0.5)$$

and an observed test statistic $x_{obs} = 60$. Values of $X$ higher than 60 give more evidence for $H_A$.


Binom(90, 0.5)
x = 60
0
25
50
75

zoom... and enhance!



The p-value is the probability above (and including) $x_{obs} = 60$ on our null distribution.

This is 0.001, which is strong evidence against the null. Chimpanzee A appears to be prosocial more than half the time.

The average prosocial rate with a partner is 0.59. Chimpanzee F has a rate of $47/90$.

Is this significantly *less* than 0.59 with $\alpha = 0.05$?

- Set up hypotheses
- Identify null distribution
- Use test statistic $x_{obs} = 47$
- Calculate p-value in the *lower* direction

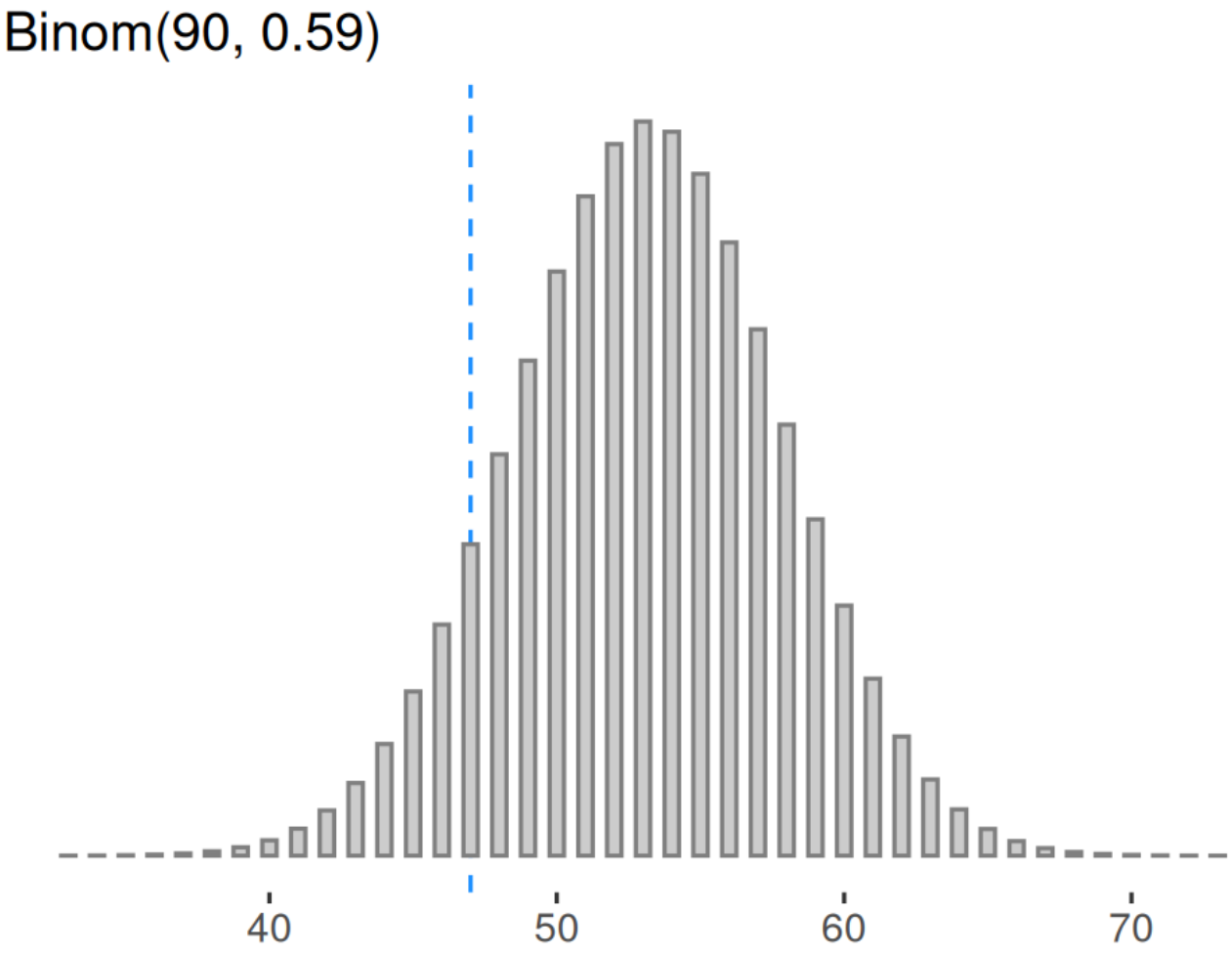
The direction of p-value calculation depends on $H_A$.

What if $H_A$ is symmetric?

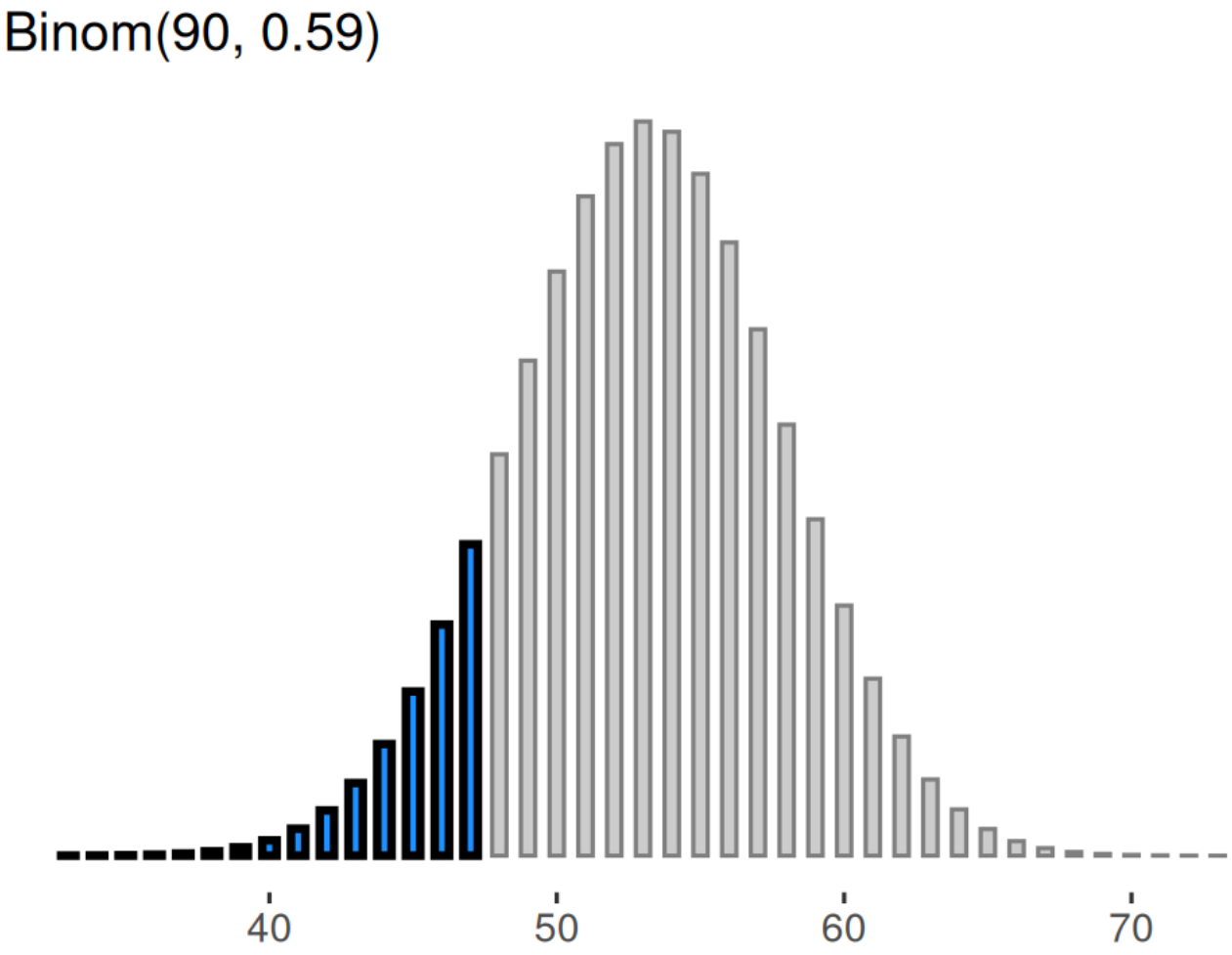
$$H_0 : p_{F,partner} = 0.59 \quad \text{versus} \quad H_A : p \neq 0.59$$

"Does chimpanzee F have a *different* prosocial rate than the average?"

# Test statistic is the same:

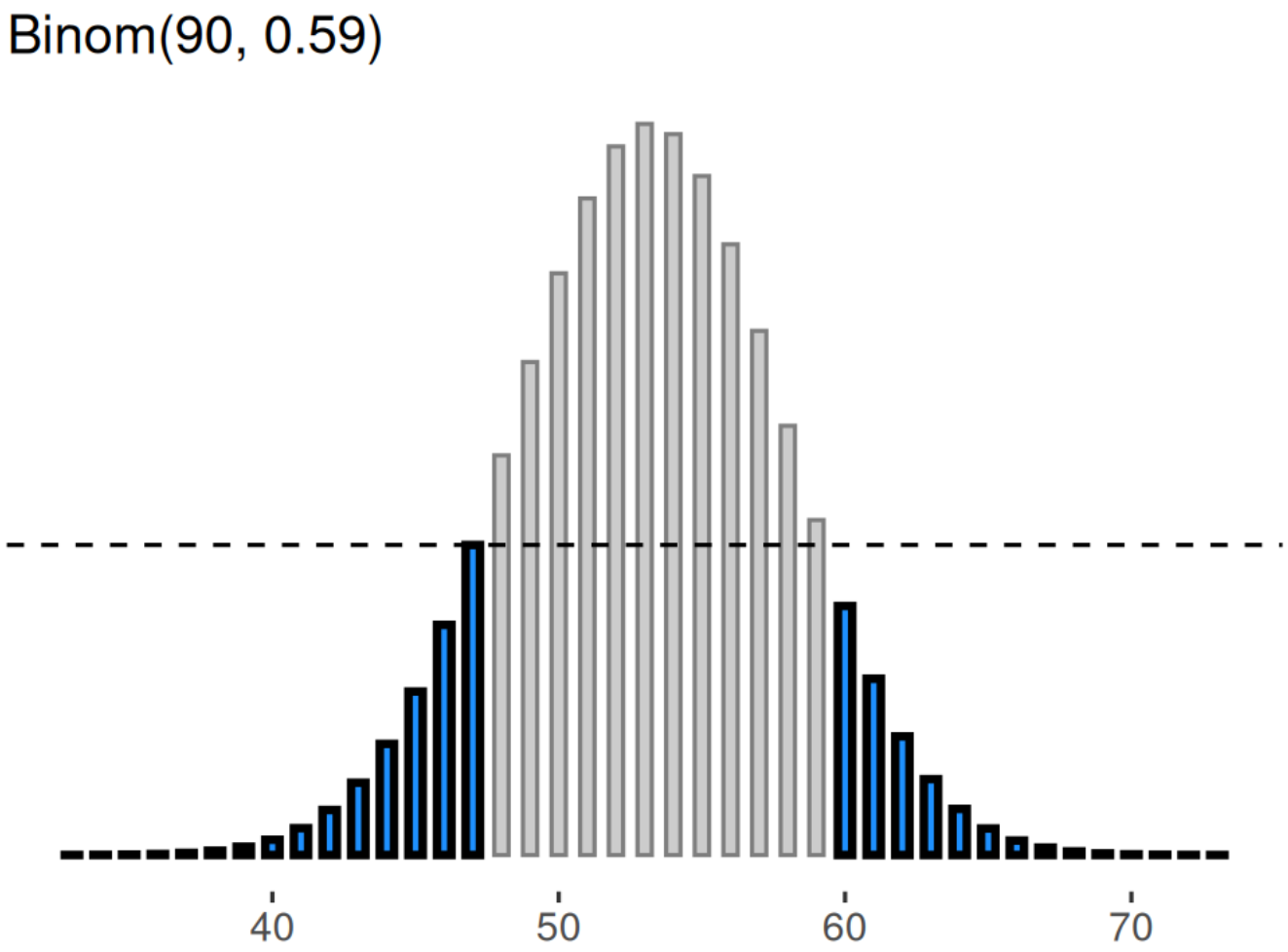


# One-sided case:



If we had two-sided hypotheses, we need to find the p-value in *both tails*.

What are the outcomes equally or less likely than $x = 47$, in either direction?




$$pval = P(X \leq 47) + P(X \geq 60)$$

In general, for different hypothesis directions:

- Test statistic does not change
- Null distribution does not change

- p-value changes
- Conclusions change

$$H_0 : p_{F,partner} = 0.59 \quad \text{versus} \quad H_A : p \neq 0.59$$

An alternative to the binomial test is a **Z test**. If the null hypothesis were true,

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

$$\hat{p} \dot{\sim} N\left(0.59, \sqrt{\frac{0.59(0.41)}{90}}\right)$$

By standardization,

$$Z = \frac{\hat{p} - 0.59}{\sqrt{\frac{0.59(0.41)}{90}}} \dot{\sim} N(0,1)$$

Our test statistic is the observed $Z$. and our null is $N(0,1)$. This works because of $n$ and $\hat{p}$.

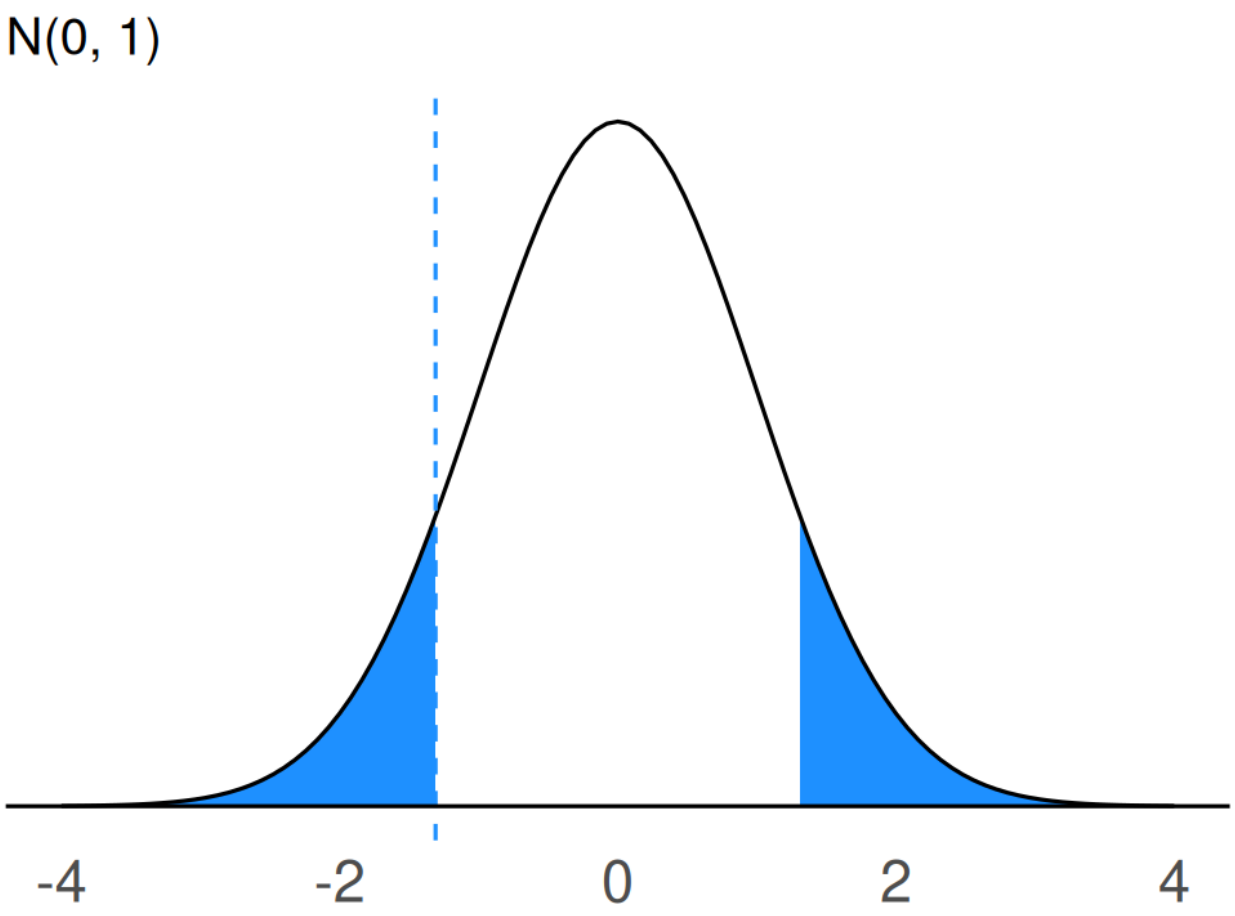
$$H_0 : p = 0.59 \quad \text{versus} \quad H_A : p \neq 0.59$$

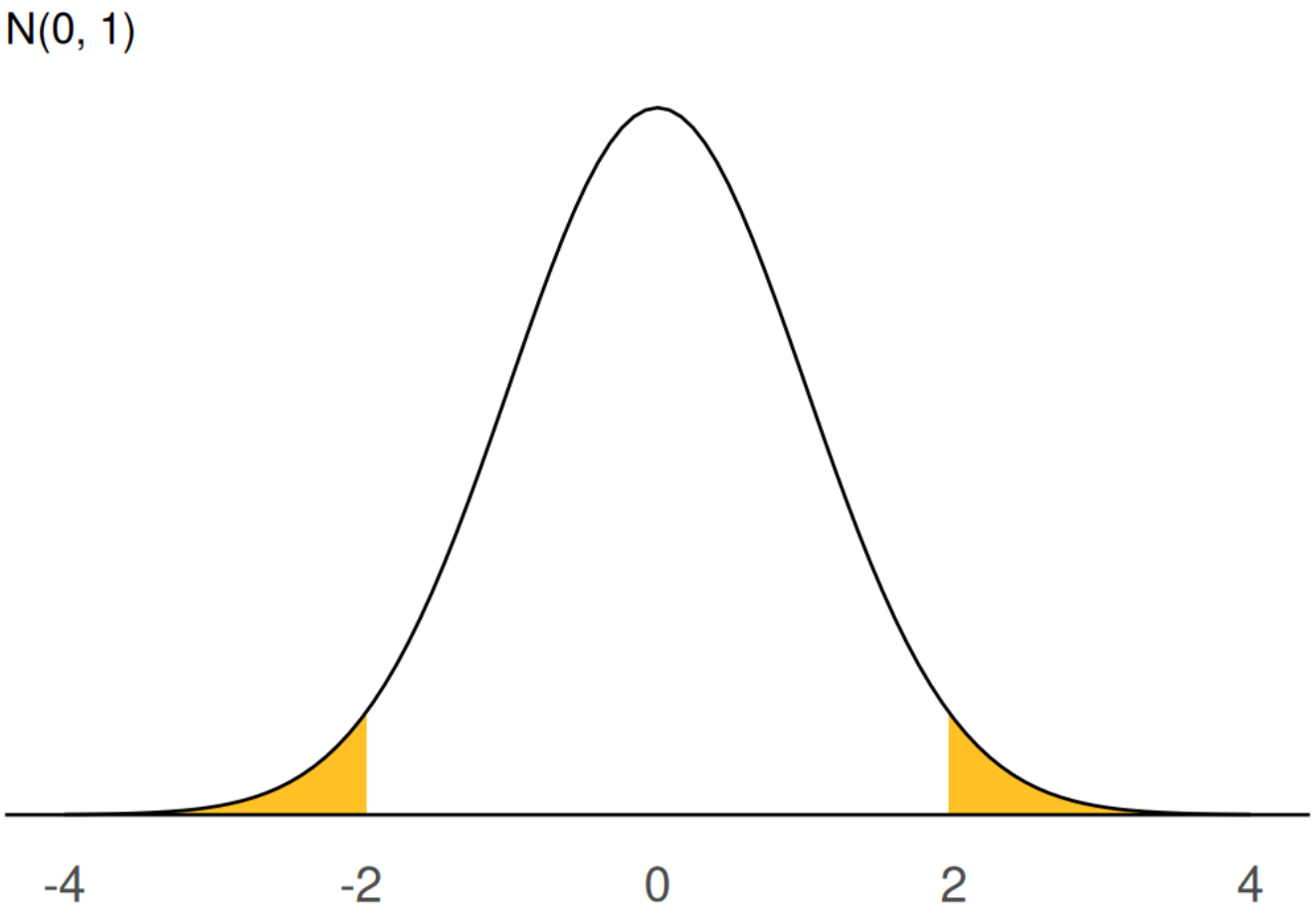
$$z_{obs} = \frac{\frac{47}{90} - 0.59}{\sqrt{\frac{0.59(0.41)}{90}}} = -1.31$$

Is -1.31 consistent with $N(0, 1)$? For a two-sided test, we need the area outside of $-1.31$ in *both tails*.

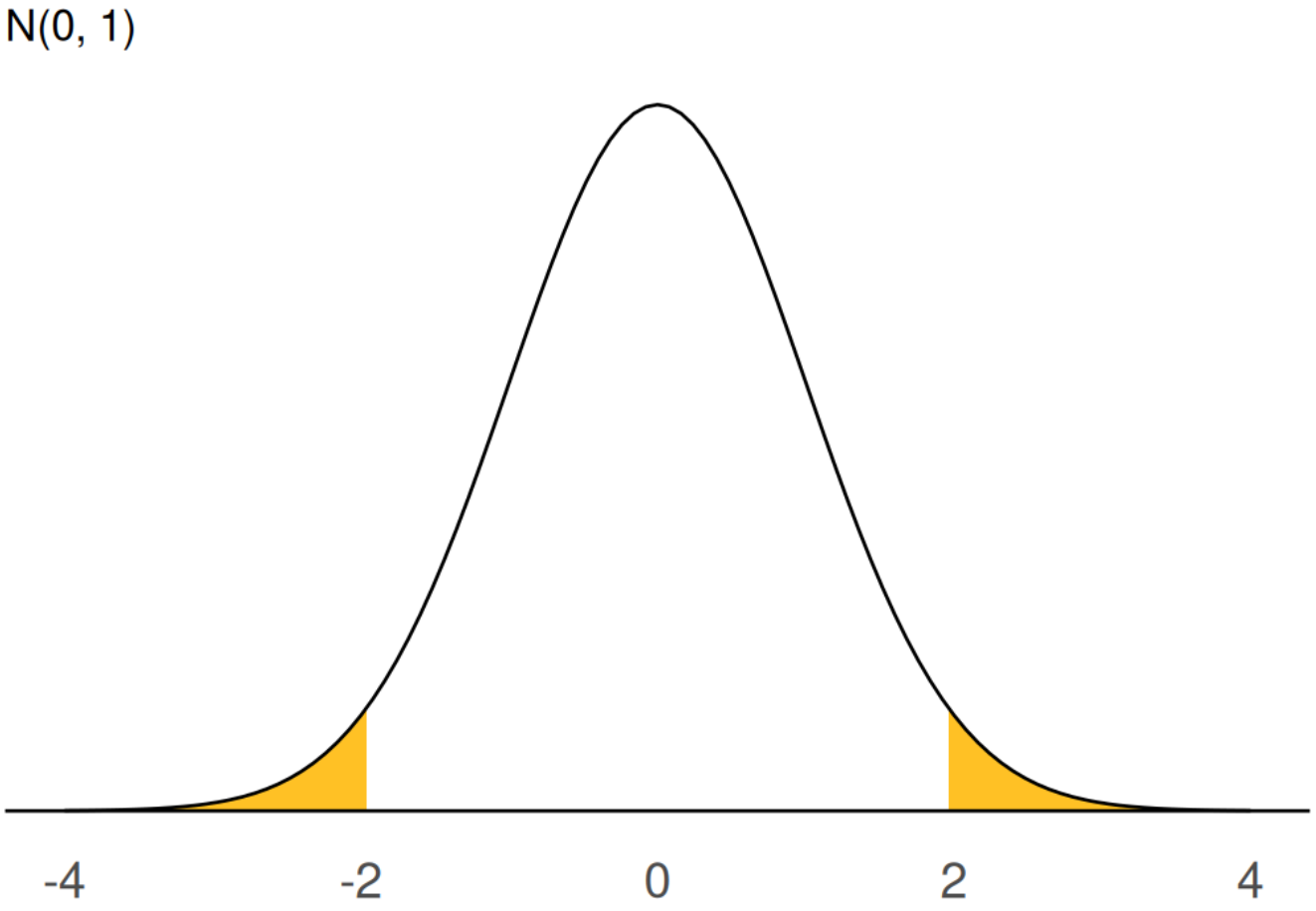
$$P(Z \leq -1.31) \; + \; P(Z \geq 1.31)$$

The p-value is 0.19, similar to the exact test.

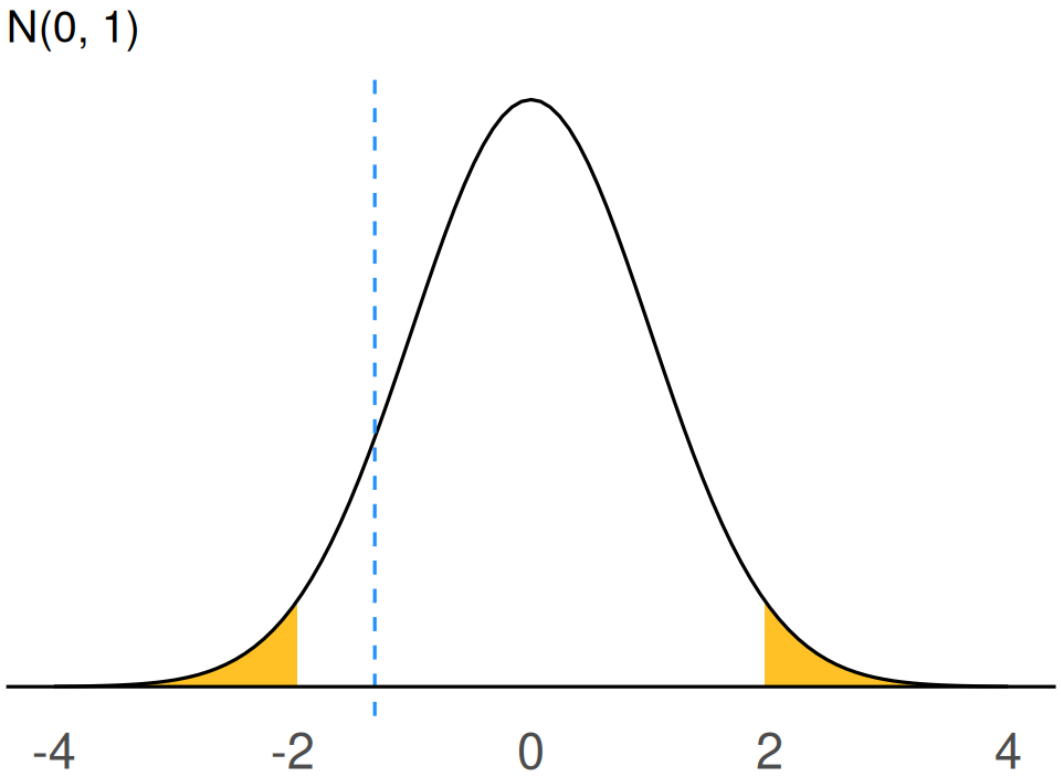

N(0, 1)
-4
-2
0
2
4




Connections to CI: Rejection region




Draw an area of size $\alpha$ in the tails.




Our test statistic is not in the rejection region.

General p-value calculation:

- If the alternative is $H_A : p < p_0$, the p-value is the area *below* the test statistic.
- If the alternative is $H_A : p > p_0$, the p-value is the area *above* the test statistic.
- If the alternative is $H_A : p \neq p_0$, the p-value is $2\times$ the area *outside* of the test statistic.